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Kernel-based calibration methods combined with multivariate feature selection to improve accuracy of near-infrared spectroscopic analysis

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ABSTRACT

We present kernel-based calibration models combined with multivariate feature selection for complex quantitative near-infrared (NIR) spectroscopic analysis of three different types of sample. Because the spectra include hundreds of features (variables), an optimal selection of features that provide relevant information for target analysis improves the accuracy of spectroscopic analysis. For this purpose, we combined feature selection with kernel partial least squares regression and kernel support vector regression (K-SVR) by evaluating ranking of the features based on their variable importance in projection scores and weight vector coefficients, respectively. Then, the methods were applied to identify components in three datasets of NIR spectra. The kernel-based models without feature selection and the kernel-based models with other feature selection methods were also used for comparison. K-SVR combined with feature selection was effective when the spectral features of samples were complex and recognition of minute spectral variation was necessary for modeling. The combination of feature selection and kernel calibration model can improve the accuracy of spectral analysis by keeping optimal features.

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1. Introduction

Near-infrared (NIR) spectroscopy is a rapid, non-destructive, multicomponent analysis method [1,2] that has been used to analyze diverse samples such as agricultural products [3,4], polymers [5–8] and petroleum/petrochemicals [9,10]. In most applications, samples are complex in composition, and NIR bands are broad due to their overtones and combination vibrations; therefore, their spectral features overlap severely and lack a distinct signature of individual components. Given these drawbacks, multivariate calibration models are used to recognize the minute spectral variations that occur due change of concentration (or property). Partial least squares regression (PLS) and its generalized version, usually kernel-based PLS regression (K-PLS), are the most commonly used methods, especially for spectroscopic analysis [11–13]. K-PLS includes a step in which a linear function is changed into one of several possible kernel functions, such as radial basis function (RBF) or a polynomial function. Alternatively, regressions based on support vector machines or on support vector regression (SVR) have become popular [14–16]. SVR can analyze high-dimensional input features efficiently to find a global solution by tuning model parameters optimally [17]. SVR can also avoid overfitting and obtain a sparse solution when the data are noisy [18,19]. SVR models are sufficiently accurate and robust to be used for analysis of chemical samples [15].

In general, spectroscopic data consist of numerous features (i.e., intensity at each wavelength) and relatively few observations (samples). In this situation, model prediction accuracy could be degraded if irrelevant features are considered in the analysis. Therefore, successful feature selection, also called "wavelength selection", can provide better prediction accuracy than using all features [20]. Feature selection can also reduce calculation time and memory requirements.

Two categories of feature selection are commonly used; these are filter and wrapper methods [21,22]. Filter methods define a feature importance metric and obtain the important features by comparison to the metric. Filter methods can be easily combined with calibration models because feature selection is performed independently of model fitting. The commonly used metrics in chemometrics are mostly based on PLS; they include variable importance in projection (VIP), loading weights, regression coefficients, and selectivity ratio (SR) [23,24]. In addition, a recently developed method, significance multivariate correlation (sMC) [25], and an extended filter method to select intervals of features instead of individual features, subset selection [26] can be found. Most filter methods based on SVR consider its weight vector coefficients [27]. On the contrary, wrapper methods such as stepwise regression, iteratively select features during optimization of a specific model's quality criteria. Many wrapper methods based on PLS have

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been presented; these include uninformative variable elimination [28], sub-window permutation analysis [29], genetic combined with PLS [30–32], competitive adaptive resampling [33], backward variable elimination PLS [34], iterative predictor weighting PLS [35], interval PLS (iPLS) [36], variations of iPLS [37,38], weighted PLS [39] and variable-weighted PLS [40] and recursive weighted PLS [41]. Some wrapper methods based on SVR have been proposed, including weighted SVR [42] and weighted least SVM [43]. In addition, the use of feature selection in the wavelet domain [44,45] and feature clustering in the network-induced supervised learning framework [46] is noticeable. Generally, filter methods are less accurate than wrapper methods but are much simpler and faster. In this paper, we focus mainly on filter methods.

We propose K-PLS and kernel-based SVR (K-SVR) combined with filter methods to improve the accuracy of NIR spectroscopic analysis. To evaluate the proposed methods, we use them to identify the specific chemical compositions detected by transmission NIR of naphtha samples, transmission NIR of etchant solutions, and diffuse reflectance NIR of apples. Using each NIR dataset, the feature selection was performed before the kernel-based modeling, and the resulting accuracies from kernel-based models with and without use of the feature selection were compared. We also used two filter methods based on PLS (i.e., SR and sMC) to evaluate the proposed method.

This paper is organized as follows. Section 2 presents the description of three datasets and two kernel-based calibration methods. Section 3 explains the details of two kernel-based calibration models combined with the multivariate feature selection. Section 3 also explains other feature selection methods to be compared with the proposed methods. Section 4 describes the application of the proposed methods for NIR spectroscopic analysis and the comparison to other feature selection methods. Section 5 concludes and offers suggestions for future research.

2. Materials and methods

2.1. Datasets

The transmission NIR dataset for naphtha consisted of 201 samples [47] that had been collected over a one-year period to maximize the compositional variation by including diverse naphtha samples from different domestic and international suppliers; as a result the chemical contents of the samples varied (Table 1). The spectral range was 4800–4000 cm^{-1} (resolution: 8 cm^{-1}). The occurrence of light scattering in the samples is insignificant and subsequent spectral preprocessing such as multiplicative scatter correction (MSC) is not required as performed in transmission NIR spectroscopic analysis of neat liquid samples in general. The baselines of spectra were linearly offset to zero at 4800 cm^{-1} for compositional analysis.

The transmission NIR dataset of etchants considered 40 solutions composed of water, H_2O_2 and HF [48] (Table 1). The spectral range was 9000–7500 cm^{-1} (resolution: 32 cm^{-1}) for the analysis.

Table 1
Description of naphtha, etchant solution and apple NIR datasets including the target properties to measure, number of spectra, number of features (variables) and ranges of property change.

Sample	Property	Unit	# of spectra	# of features	Property range	
					Min	Max
Naphtha	Total paraffin	wt%	201	400	75.65	87.16
	Total naphthene				9.49	15.61
	Total aromatic				2.95	7.76
Etchant solution	Water	wt%	40	98	72.82	93.48
	H_2O_2				5.91	25.19
	HF				0.04	2.18
Apple	Sugar content	Brix%	300	576	7.55	18.40

The third dataset consisted of diffuse reflectance NIR spectra of 150 apples cultivated in diverse areas in Korea. To acquire the spectra, each apple was subjected to NIR/VIS radiation at room temperature (~25 °C), and diffusely reflected radiation was collected using an integrating sphere (Cary 5G, Varian, Santa Clara, CA, USA). For each apple, two spectra were collected in the 500–1650 nm range (resolution: 2 nm) from opposite sides. All of the spectra were subject to multiplicative scatter correction (MSC) before analysis. Sugar content (Table 1) of the samples was measured using a hand-held sugar-refractometer (PR 32, Atago, Japan).

2.2. Multivariate methods

2.2.1. Kernel partial least squares regression

In many areas of research and industrial situations data can exhibit nonlinear behavior. A common approach to modeling nonlinear relations is the K-PLS [49,50] approach, which maps the original input spectra onto a higher-dimensional basis function space $\mathscr{F}$, then applies the PLS algorithm to the transformed spectra. $\mathscr{F}$ captures a rich structure and thus is more suitable for some chemometrics regression problems than the original input space. However, $\mathscr{F}$ can be of high and even infinite dimension; which this happens, PLS regression is computationally burdensome. This problem can be solved using the kernel trick: the kernel function evaluates an inner product between two vectors in $\mathscr{F}$; this process provides a similarity measure $k(x_i, x_j) = \Phi(x_i)^T\Phi(x_j)$, $\forall\ x_i, x_j \in X$ between pairs of spectra x_i and x_j. Using the kernel trick avoids having to explicitly construct $\mathscr{F}$ [12], and estimating the PLS in a feature space only requires linear algebra that is as simple as in linear PLS. Detailed formalism, equations and algorithms have been presented earlier [12,50].

2.2.2. Kernel support vector regression

The support vector machine (SVM) was invented to solve the problem of classifying two classes by finding two parallel (p-1)-dimensional hyperplanes that maximize the distance (or margin) between the classes in p-dimensional space, where p is the number of features. The hyperplane placed midway between the two parallel hyperplanes serves as a discriminant function [51] (Fig. 1a [52]). SVR has a modified constraint to minimize the error between the estimated value and the target value. Vapnik's ϵ-insensitive loss function only penalizes predicted values that deviate by $>\epsilon$ from the actual value (Fig. 1b). Observations that have prediction errors $>\epsilon$ are called support vectors and only these vectors determine the final SVR model (Fig. 1). The values of ϵ and a cost parameter C are determined depending on the problem and data. The value of ϵ can affect the number of support vectors used to construct the regression function. Choice of ϵ is difficult in real applications, so ν-SVR [53,54] is commonly used instead, because it allows the user to specify the parameter ($\nu \in (0, 1]$) easily in a relatively narrow range. K-SVR considers nonlinearity of the data based on the same principle of K-PLS by introducing a kernel function, due to the fact that only the inner product is used in all calculations [55,56]. Detailed theoretical background on SVM and SVR has been provided elsewhere [55–57].

3. Kernel-based models combined with feature selection

Many feature selection methods rank the features ranking rather providing an explicit best subset. We apply a feature selection method that uses feature ranking and embed it each of the two kernel models, because it supplies the relevance of each feature to the target analysis; this attribute allows control of the number of selected features. As a ranking criterion, we use the VIP score of linear PLS and the weight vector of linear SVR because of their compatibility with kernel methods. Each step in the analysis halves the remaining number of features (m) (i.e., $m \in \{p, p/2, p/2^2, \ldots\}$) because spectroscopic data are highly correlated each other, and include some noise features. Then we optimize

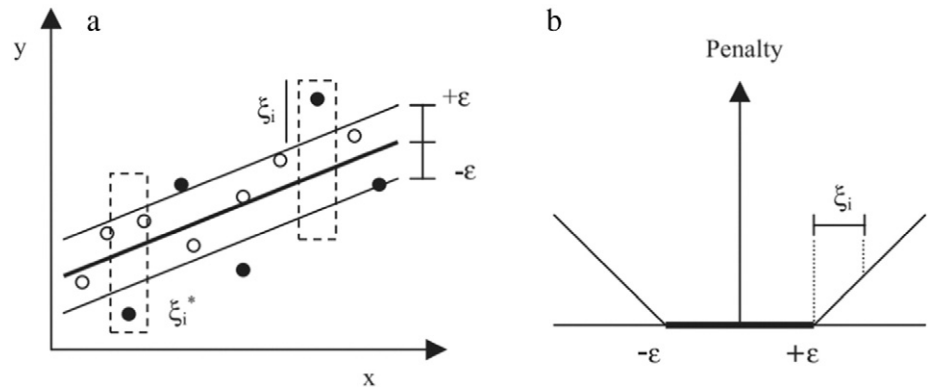

Fig. 1. The parameter setting for linear SVR. (a) ξ_i^* represents an error > tolerance ϵ; (b) the cost parameter C determines the slopes of the Vapnik's ϵ-insensitive loss function.

kernel models by re-determining the parameter(s) for the reduced input features. We repeat the procedure until the number of remaining features is relatively small (~10). Finally we determine the best model with the lowest error among models with different $m \in \{p, p/2, p/2^2, \ldots\}$.

We first select features, then construct the calibration model because the two steps can work independently. Importance features can be more effectively captured in the linear space than in the nonlinear space because the linear space keeps the properties of the original input features. Then we can explore a dynamic nonlinear behavior with those selected features for better prediction accuracy.

In this study, we consider only the radial basis function (RBF) kernel

$$k(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right), \tag{1}$$

where γ is the kernel parameter that controls the width of the Gaussian distribution. The RBF kernel can cover general input space, and the polynomial kernel can be regarded as a special case of RBF kernel. The RBF kernel's stability and accuracy have been demonstrated for many cases [58,59].

3.1. Kernel PLS combined with feature selection

First, we rank all features in descending order of VIP scores which have been used extensively in chemometrics as the most popular filter method based on linear PLS [60]. The VIP score is a measure of the importance of the jth feature which reflects the variability of each latent component and the relative weight of each component assigned to the jth feature. The VIP measure ν_j is defined as [61]

$$\nu_j = \sqrt{p / SSR \sum_{a=1}^{A} SSR_a \left(w_{aj} / \|w_a\|\right)^2}, \quad j = 1, \ldots, p, \tag{2}$$

where p is the number of features, SSR_a is the variability (sum of squares) explained by the ath component, SSR is the total variability, w_a is the weight vector of component a, and w_{aj} is the weight of component a assigned to feature j. ν_j measures the importance of feature j and quantifies the variability of each latent component and the relative weight of each component assigned to that feature. Hence, ν_j is a measure of the contribution of feature j according to the variance explained by each PLS component, where $(w_{aj}/\|w_a\|)^2$ represents the relative weight feature j to the component a. We compare values of ν_j as a ranking criterion in our K-PLS method when eliminating features.

The proposed algorithm (K-PLS-VIP) consists of three steps (Algorithm 1). Steps 1 and 2 sort the features by ν_j scores obtained using a trained linear PLS. Step 3 consists of an outer loop that eliminates features, and an inner loop that optimizes the K-PLS for the remaining feature subset. In this algorithm, the parameter γ for the RBF kernel has the range $-15 \le \log_2\gamma \le 15$ in increments of 1. The search range of γ should be wide enough because the optimal γ (Eq. (1)) affects the prediction accuracy directly, and must be found in either dataset.

Algorithm 1. Kernel PLS embedded with feature selection

Input: Training sets, (x_i, y_i), $i = 1, \ldots, l$.
Output: Best performance for each feature size

Step 1. Train a linear PLS model under the fixed number of components (which depends on the data).
Step 2. Sort the features according to the VIP scores calculated by the linear PLS model.
Step 3. (outer loop) For each feature size $m \in \{p, \frac{p}{2}, \frac{p}{2^2}, \ldots\}$,

(a) Construct the training set that has only the first m features in the sorted feature list, f_j, $j = 1, \ldots, m$ which are a nested subset of whole features.
(b) (inner loop) For each parameter $\gamma \in \{2^{-15}, 2^{-14}, \ldots, 2^{14}, 2^{15}\}$.

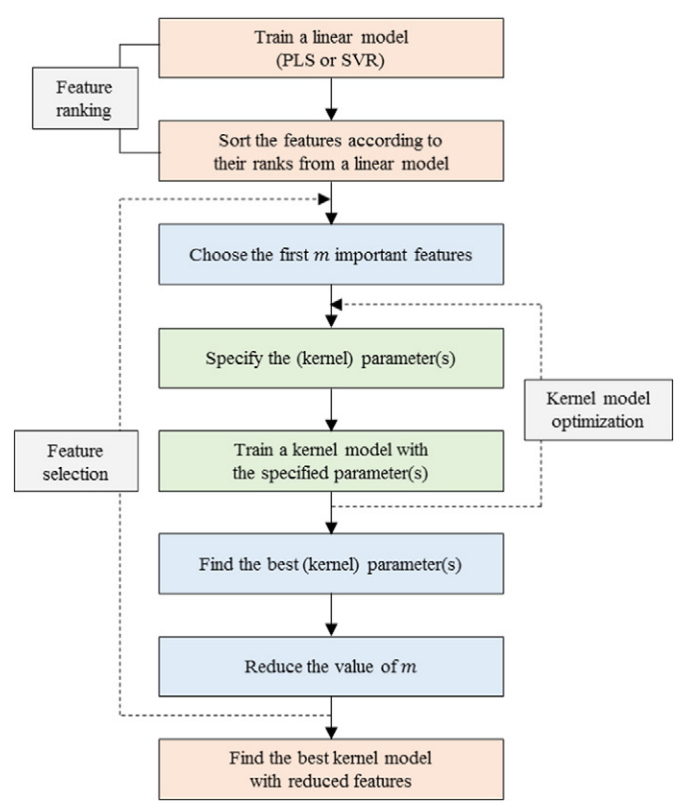

Fig. 2. Flow chart of the algorithms.

Table 2
SEPs (std.) resulted for measurements of total paraffin, naphthene and aromatic concentrations of naphtha samples using five methods. The numbers in brackets indicate the determined optimal number of features.

Property	Method				Unit: wt%
	PLS	K-PLS	K-SVR	K-PLS-VIP	K-SVR-WV
Total paraffin	0.78 (0.02)	0.73 (0.01)	0.73 (0.02)	0.72 (0.01) [200]	0.68 (0.02) [200]
Total naphthene	0.39 (0.01)	0.36 (0.01)	0.36 (0.01)	0.35 (0.02) [200]	0.35 (0.01) [200]
Total aromatic	0.39 (0.01)	0.38 (0.01)	0.39 (0.01)	0.38 (0.01) [All]	0.37 (0.01) [200]

(1) Train kernel PLS on the constructed training set with a parameter γ.
(2) Predict the testing set using the model.
(3) Record training error and SEP.

(c) Record the lowest SEP.

3.2. Kernel SVR combined with feature selection

In a linear SVM model, a weight vector can be used to identify the importance of each input feature [62], i.e., a large magnitude of weight vector coefficient for jth feature implies that this feature has a strong influence when ranking the features. Because only the weight vector in a linear SVM model has this characteristic, we use the linear SVM model [63] and rank all features by the magnitudes of weight vector coefficients. This process is based on the fact that the optimal hyperplane is the same for the deletion of jth feature if the jth coefficient of the weight vector is zero. Even if the jth coefficient is not zero, the deletion of jth feature has little effect on the optimal hyperplane if the magnitude of coefficient is small, so features with small weight vector coefficients can be deleted [64].

After obtaining the ranking list, we used a grid search to optimize K-SVR to find the best kernel parameter γ in the RBF kernel and the implicit parameter v in SVR. To reduce complexity, we fixed the cost parameter C in SVR in advance as the difference between the maximum and minimum of the target (property) data.

We also suggest an alternative algorithm (K-SVR-WV) (Algorithm 2). It is designed to find γ more precisely than v because we found empirically that γ influences prediction accuracy more than does v. Except for adding the parameter v in the inner loop, the procedure is the same as K-PLS with feature selection. Both algorithms follow the same procedure (Fig. 2).

Algorithm 2. Kernel SVR embedded with feature selection

Input: Training sets, (x_i, y_i), $i = 1, \ldots, l$.

Output: Best performance for each feature size

Step 1. Train a linear SVR model under the fixed value of v and C (which depends on the data).
Step 2. Sort the features according to the magnitudes of weight vector coefficients obtained by the linear SVR model.
Step 3. (outer loop) For each feature size $m \in \{p, \frac{p}{2}, \frac{p}{2^2}, \ldots\}$,
(a) Construct the training set that has only the first m features in the sorted feature list, f_j, $j = 1, \ldots, m$ which are a nested subset of whole features.
(b) (inner loop) For each parameter $\nu \in \{0.1, 0.5, 0.9\}$ and $\gamma \in \{2^{-15}, 2^{-14}, \ldots, 2^{14}, 2^{15}\}$.
(1) Train kernel SVR on the constructed training set with parameters v and γ.
(2) Predict the testing set using the model.
(3) Record training error and SEP.

(c) Record the lowest SEP.

3.3. Other feature selection methods: selectivity ratio and significance multivariate correlation

We compare our proposed methods with SR and sMC, which have their own thresholds to select important variables [65,25]. However for fair comparison, we will treat SR and sMC as a ranking method like VIP and weight vector.

3.3.1. Selectivity ratio

SR is the ratio between the explained variance (SSR) and the residual variance (SSE) [23,24]; given the PLS regression vector $\hat{\beta}_{PLS}$, target projection is performed by projecting the rows of input X onto the normalized regression vector

$$t_{TP} = X\hat{\beta}_{PLS} / \left\| \hat{\beta}_{PLS} \right\|; \quad (3)$$

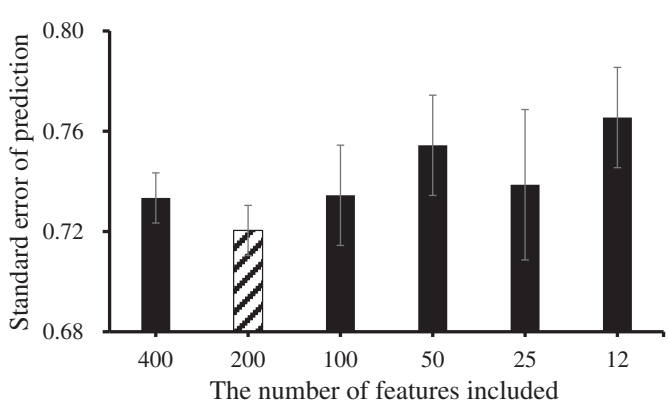

Fig. 3. SEPs with standard deviations during the procedure of K-PLS-VIP for Paraffin. The bar marked with slanted line indicates the best model with the lowest SEP.

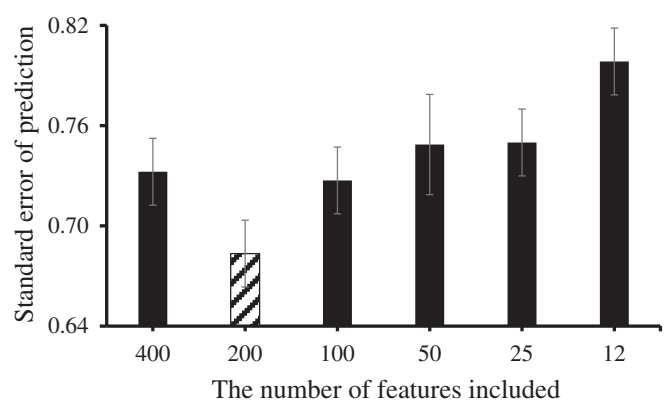

Fig. 4. SEPs with standard deviations during the procedure of K-SVR-WV for Paraffin. The bar marked with slanted line indicates the best model with the lowest SEP.

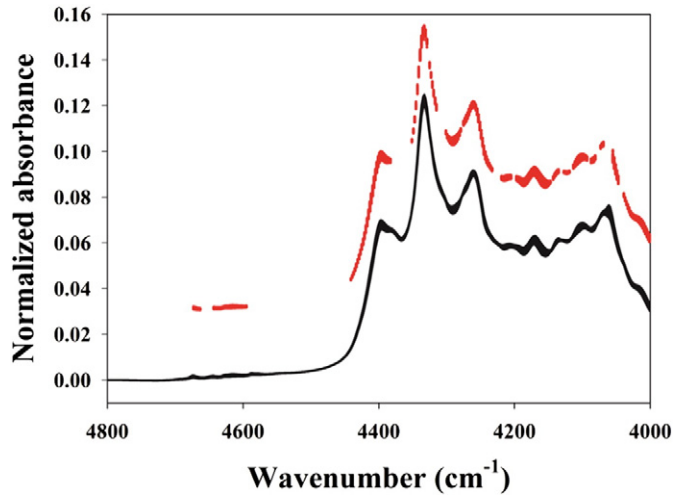

Fig. 5. Normalized NIR spectra of all naphtha samples (black) and 200 optimally selected features in K-SVR-WV by ranking the magnitudes of weight vector coefficients to determine total paraffin concentration (red). The selected features are offset for clarity.

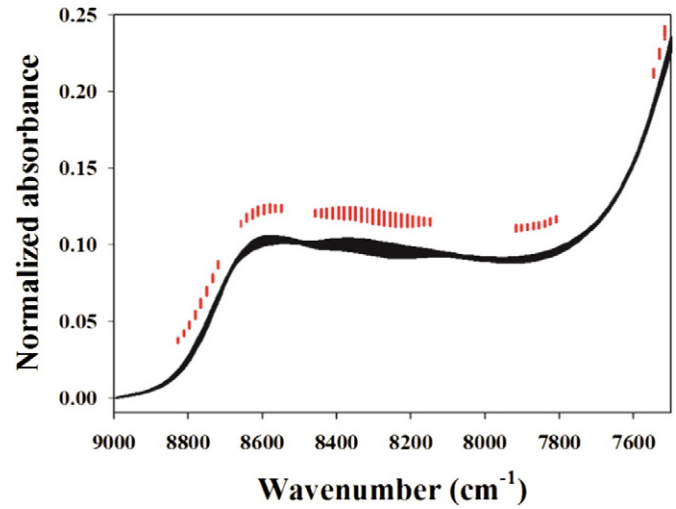

Fig. 6. Normalized NIR spectra of etchant solutions (black) and 49 optimally selected features in K-SVR-WV by ranking the magnitudes of weight vector coefficients for the determination of water concentration (red). The selected features are offset for the clarity.

as a result, t_{TP} is proportional to the predicted values $\hat{y}$. A loading vector, p_{TP}, is acquired via the projection of the columns of X onto the score vector, t_{TP},

$$p_{TP} = X' t_{TP} / (t_{TP}' t_{TP}). \tag{4}$$

To determine the importance of the variable, SR uses the ratio

$$SR_j = SSR_j / SSE_j \tag{5}$$

of the explained variance

$$SSR_j = \left\| t_{TP} p_{TP_j}' \right\|^2 \tag{6}$$

to the residual variance

$$SSE_j = \left\| e_{TP_j} \right\|^2. \tag{7}$$

for each variable j.

3.3.2. Significance multivariate correlation

SMC estimates the actual sources of the variation of features from the PLS regression (i.e. SSR and SSE), and uses them to determine the importance of features with respect to the regression model. Unlike SR, SMC removes the procedure of the projection in Eq. (4) to avoid the influence of irrelevant information contained in X. Therefore, the normalized regression vector as shown in Eq. (8) as a loading vector.

$$p_{sMC} = \hat{\beta}_{PLS} / \left\| \hat{\beta}_{PLS} \right\| \tag{8}$$

$$t_{sMC} = X p_{sMC} = X \hat{\beta}_{PLS} / \left\| \hat{\beta}_{PLS} \right\| = \hat{y} / \left\| \hat{\beta}_{PLS} \right\|, \tag{9}$$

$$X = t_{sMC} p_{sMC}' + E_{sMC} = \left(\hat{y} \hat{\beta}_{PLS}' \right) / \left\| \hat{\beta}_{PLS} \right\|^2 + E_{sMC}, \tag{10}$$

$$x_j = t_{sMC} p_{sMC_j}' + e_{sMC_j} = \frac{\hat{y} \hat{\beta}_{PLS_j}}{\left\| \hat{\beta}_{PLS_j} \right\|^2} + e_{sMC\,j}. \tag{11}$$

sMC is not a complete basic rotation and the SSR of jth feature

$$SSR_j = \left\| t_{sMC} p_{sMC_j}' \right\|^2 = \left\| \hat{y} \hat{\beta}_{PLS_j}' \Big/ \left\| \hat{\beta}_{PLS_j} \right\|^2 \right\|^2 \tag{12}$$

might not be orthogonal to the SSE of jth feature

$$SSE_j = \left\| e_{sMC_j} \right\|^2 = \left\| x_j - \hat{y} \hat{\beta}_{PLS_j}' \Big/ \left\| \hat{\beta}_{PLS_j} \right\|^2 \right\|^2, \tag{13}$$

but SSR_j quantifies the relevant variation in the predicted values projected back onto the original X by the PLS regression vector as

$$MSR_j = SSR_j / 1,\ MSE_j = SSE_j / (n-2), \tag{14}$$

$$sMC_j = MSR_j / MSE_j, \tag{15}$$

where n is the sample size.

Table 3
SEPs (std.) resulted for measurements of total paraffin, naphthene and aromatic concentrations of naphtha samples using six methods. The numbers in brackets indicate the determined optimal number of features.

Property	Method					Unit: wt%
	K-PLS-SR	K-PLS-sMC	K-PLS-VIP	K-SVR-SR	K-SVR-sMC	K-SVR-WV
Total paraffin	0.73 (0.01) [200]	0.73 (0.01) [200]	0.72 (0.01) [200]	0.73 (0.02) [All]	0.73 (0.02) [All]	0.68 (0.02) [200]
Total naphthene	0.36 (0.02) [200]	0.36 (0.02) [200]	0.35 (0.02) [200]	0.36 (0.01) [All]	0.36 (0.01) [All]	0.35 (0.01) [200]
Total aromatic	0.39 (0.01) [All]	0.38 (0.02) [100]	0.38 (0.01) [All]	0.39 (0.01) [All]	0.39 (0.01) [All]	0.37 (0.01) [200]

Table 4
SEPs (std.) resulted for measurements of water, H_2O_2 and HF concentrations of etchant solutions using five methods. The numbers in brackets indicate the determined optimal number of features.

Property	PLS	K-PLS	K-SVR	K-PLS-VIP	Unit: wt% K-SVR-WV
Water	0.73 (0.05)	0.54 (0.03)	0.54 (0.03)	0.54 (0.03) [All]	0.48 (0.05) [49]
H_2O_2	0.76 (0.03)	0.62 (0.02)	0.60 (0.02)	0.54 (0.02) [All]	0.51 (0.03) [24]
HF	0.18 (0.02)	0.15 (0.01)	0.13 (0.01)	0.13 (0.01) [24]	0.12 (0.01) [49]

Table 5
SEPs (std.) resulted for measurements of water, H_2O_2 and HF concentrations of etchant solutions using six methods. The numbers in brackets indicate the determined optimal number of features.

Property	Method					Unit: wt%
	K-PLS-SR	K-PLS-sMC	K-PLS-VIP	K-SVR-SR	K-SVR-sMC	K-SVR-WV
Water	0.54 (0.03) [All]	0.48 (0.04) [49]	0.54 (0.03) [All]	0.54 (0.03) [All]	0.54 (0.03) [All]	0.48 (0.05) [49]
H_2O_2	0.60 (0.03) [49]	0.61 (0.03) [49]	0.54 (0.02) [All]	0.60 (0.02) [All]	0.60 (0.02) [All]	0.51 (0.03) [24]
HF	0.14 (0.01) [49]	0.14 (0.01) [49]	0.13 (0.01) [24]	0.13 (0.01) [All]	0.13 (0.01) [All]	0.12 (0.01) [49]

4. Results and discussion

To analyze each NIR dataset, we first evaluated PLS, K-PLS and K-SVR without feature selection. We also evaluated four models with other feature selection methods, K-PLS-SR, K-PLS-sMC, K-SVR-SR, and K-SVR-sMC. Then we compared the results respectively to those obtained using proposed methods, K-PLS-VIP and K-SVR-WV. We calculated a standard error of prediction (SEP) for each case as a performance measure. To avoid any bias between the calibration and validation data, 30 random repetitions of 5-fold cross validation were performed. The average and standard deviation of SEPs over 150 runs were reported. All algorithms were run in MATLAB R2013a (The MathWorks Inc.).

4.1. Determination of compositions for naphtha samples

PLS was initially used to determine the total paraffin, naphthene and aromatic concentrations of the naphtha samples using NIR spectroscopy. The chemical composition of naphtha is complex [47,66], so its NIR spectral feature is highly overlapped with those of individual components [47,66,67]. Therefore, the number components (factors, latent variables) must be optimized to avoid overfitting under when analyzing naphtha NIR spectra. For this purpose, 10 components that explained ~90% of the total variance were used in PLS.

SEPs (Table 2) were obtained after using PLS, K-PLS and K-SVR to measure the total paraffin, naphthene and aromatic concentrations. For all three properties, the value of that gave the lowest error was 2^{13}. For the total paraffin and naphthene concentrations, the kernel-based models decreased the errors slightly compared to PLS. This result suggests that the feature space mapped by the RBF kernel is more effective than the original input space in recognizing relevant spectral variations in complex naphtha spectra. The SEPs of K-PLS and K-SVR did not differ. For the total aromatic concentration, the kernel-based models did not improve the prediction accuracy. Because the NIR spectral features of aromatic compounds are more distinct than those of paraffin and naphthene, and because the aromatic bands in the 4700–4500 cm^{-1} range do not overlap those of the paraffin and naphthene compounds, the relevant spectral variation can be recognized relatively easily; therefore use of the kernel-based models is not advantageous in this case.

Next, K-PLS-VIP and K-SVR-WV were also tested for the same analysis by determining the optimal number of features for each case (Table 2). Use of K-SVR-WV decreased the SEP. As a representative example, Figs. 3 and 4 for paraffin show the process to determine the best model and this process was applied at the same way for other properties. NIR spectral features of paraffin compounds are similar due to their similarity in molecular structure [47].The result suggests that the differentiation of the similar spectral features could be enhanced by the filtering effect by which feature ranking removes features that have relatively low importance. For the total naphthene and aromatic concentrations, the use of K-SVR-WV did not improve the accuracy, because the naphthene and aromatic compounds have more distinct spectral bands than do paraffin compounds [68,69].

K-SVR-WV selected 200 optimal features (Fig. 5) by ranking the magnitudes of weight vector coefficients to determine total paraffin concentration. Feature selection tended to keep the original global

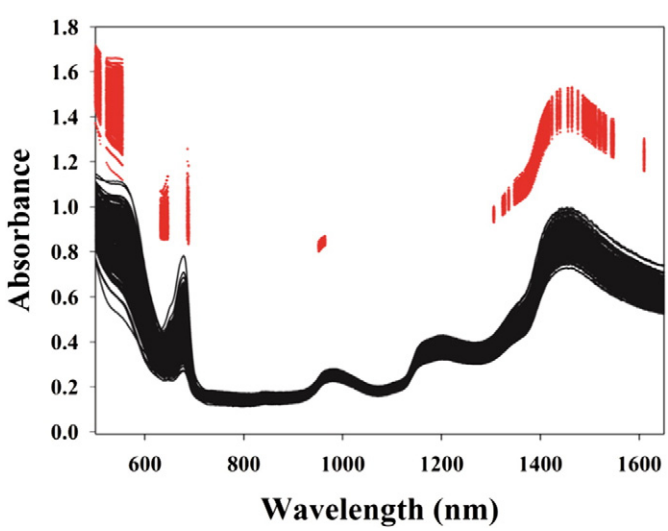

Fig. 7. MSC-treated diffuse reflectance spectra of apples (black) and 114 optimally selected features in K-SVR-WV by ranking the magnitudes of weight vector coefficients for the determination of sugar content (red). The selected features are offset for the clarity.

Table 6
SEPs (std.) resulted for measurements of sugar content in apples using five methods. The numbers in brackets indicate the determined optimal number of features.

Property	Method				Unit: Brix%
	PLS	K-PLS	K-SVR	K-PLS-VIP	K-SVR-WV
Sugar content	1.32 (0.01)	1.25 (0.01)	1.25 (0.02)	1.23 (0.03) [228]	1.18 (0.03) [114]

Table 7
SEPs (std.) resulted for measurements of sugar content of apples using six methods. The numbers in brackets indicate the determined optimal number of features.

Property	K-PLS-SR	K-PLS-sMC	K-PLS-VIP	K-SVR-SR	K-SVR-sMC	Unit: Brix% K-SVR-WV
Sugar content	1.24 (0.03) [114]	1.23 (0.04) [57]	1.23 (0.03) [228]	1.25 (0.02) [All]	1.25 (0.02) [All]	1.18 (0.03) [114]

features by including the major absorption bands and simultaneously excluding redundant features. This result indicates that feature selection based on weight vector coefficients is well adapted to K-SVR, especially when measuring the concentration of compounds with indistinct and overlapping spectral features.

SEPs (Table 3) of K-PLS-SR, K-PLS-sMC, K-SVR-SR, and K-SVR-sMC were obtained for naphtha samples. In this comparison, the most effective combination was K-SVR-WV. sMC and SR were effective in K-PLS but had no effect in K-SVR, possibly because these methods are derived from PLS. None of the feature selection methods applied to K-PLS was obviously dominant. Therefore, we conclude that K-PLS had an advantage of feature selection when using a feature selection method based on PLS, and that K-SVR had an advantage of feature selection when using feature selection method based on an SVR.

4.2. Determination of concentrations for etchant solutions

NIR spectra (Fig. 6) obtained from etchant solutions (bottom) in the 9000–7500 cm^{-1} range included superposed OH bands at ~8450 cm^{-1}. Because the concentration changes of H_2O_2 and water in the samples changed more than did the concentration of HF, the spectral variations were more noticeable in the H_2O_2 (8620 cm^{-1}) and water (8390 cm^{-1}) bands than in the HF band, for which the relevant spectral feature is difficult to recognize because they vary little due to the low concentrations of HF. Overall, the bands of each component were broad and overlapped widely with each other. Therefore, a method that can effectively recognize component-specific spectral information would increase the accuracy with which concentration is determined.

SEPs (Table 4) resulting from determining the concentrations of water, H_2O_2 and HF using PLS, K-PLS, K-SVR, K-PLS-VIP and K-SVR-WV were obtained with the number of components in PLS and K-PLS set to 2; the optimal γ was 2^{12} for all three components. Compared with PLS, SEPs were decreased by combining the kernel with either PLS and SVR. The kernel mapping effectively extracted information to correlate with the concentration variations of each component. The improvement of accuracy is relatively more significant for the determination of water concentrations with the use of K-SVR-WV than with the use of the other methods. The water band is perturbed by the presence of both components due to their interaction with water vibrations. Under this situation, appropriate selection of features while minimizing the influence from the other two components is a difficult task. This difficulty could explain why K-SVR-WV had higher accuracy in the measurement of water concentrations than did the other methods. Forty-nine features were selected (Fig. 6); they were widely spread around the water absorption band at 8450 cm^{-1}. K-SVR-WV could also find information relevant to HF that was present at low concentrations and that had indistinct spectral features.

As in the naphtha samples, K-SVR-WV was more accurate than the other methods, and sMC and SR had no effect in K-SVR for the same reason as for the naphtha samples (Table 5). Among feature selection methods applied to K-PLS, K-PLS-VIP obtained the lowest SEP at H_2O_2 and HF, and even with significant difference at H_2O_2.

4.3. Determination of sugar content in apples

MSC-treated diffuse reflectance spectra (Fig. 7) of the apples were broad due to the presence of water NIR bands, whereas the spectral features related to sugars were minute and obscured by the water bands [70,71]. Therefore, a calibration method able to perceive the hidden features under the big water bands is beneficial. The number of components in PLS and K-PLS was chosen as 5, because SEP was smallest at this value.

SEPs (Table 6) were obtained after determining the sugar content using PLS, K-PLS, K-SVR, K-PLS-VIP and K-SVR-WV. The optimal γ was 2^{-4}. The use of K-SVR-WV improved accuracy by selecting only 100 out of 576 features; most of the selected features were near the major bands at 535, 640, and 1390 nm (Fig. 7). Due to the high absorbance, these bands contain spectral information pertinent to sugar content, so selection of features near them is reasonable. We could decrease the error by keeping the optimal number of features to describe the variation of sugar content while simultaneously eliminating redundant features. The comparison result for apples was consistent with those for naphtha samples and etchant solutions (Table 7).

5. Conclusion

Multivariate feature selection was combined with K-PLS and K-SVR. The combination was evaluated as successful by comparing the ranking of features based on their VIP scores and weight vector coefficients, respectively. The consistency between feature selection and kernel calibration model was proven to be an important problem for the accuracy. Application of the proposed methods for NIR spectroscopic analysis of naphtha, etchant solution and apple samples revealed that K-SVR-WV was effective to improve the accuracies of their quantitative analyses. The results indicate that K-SVR-WV was superior when the spectral features of samples were highly complex and recognition of minute spectral variation was desired. In these cases, the feature selection for analysis kept only the features that provided useful descriptive information.

SVR is widely used for chemometric analysis, so this combination of K-SVR and feature selection may be a meaningful improvement in the method, K-SVR-WV. In future work we will increase the diversity of sample datasets to include those measured using other methods such as Raman and NMR spectroscopy, and will analyze how the accuracy of K-SVR-WV varies in relation to data types (or spectral selectivity) and to noise level in these data.

An application of a Markov blanket feature selection method such as in [72] coud be an interesting future research topic.

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